

Rerandomization and Regression Adjustment

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Spring 2026

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- How should we use general covariates?
- Design stage: rerandomization
- Analysis stage: covariate adjustment

Covariate balance

- CRE with $n \times K$ covariate matrix $\mathbf{X} = (X_1, \dots, X_n)^\top$
 - center $\mathbf{X}$ for simplicity: $\bar{X} = n^{-1} \sum_{i=1}^n X_i = 0$

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$$\hat{\tau}_X = n_1^{-1} \sum_{i=1}^n Z_i X_i - n_0^{-1} \sum_{i=1}^n (1 - Z_i) X_i$$

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- CRE balances covariates: $\mathbb{E}(\hat{\tau}_X) = 0$
- Undesired covariate imbalance may appear in realized treatment allocation $\rightsquigarrow$ Mahalanobis distance measure for covariate balance

Mahalanobis distance measure

- $M = \hat{\tau}_X^\top \text{cov}(\hat{\tau}_X)^{-1} \hat{\tau}_X$

$$\text{cov}(\hat{\tau}_X) = \frac{n}{n_1 n_0} S_X^2, \quad S_X^2 = (n-1)^{-1} \sum_{i=1}^n X_i X_i^\top$$

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- Use of M

- Check the quality of randomization
- Rerandomization

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 - $a = 0$: very few feasible treatment allocations

Rerandomization

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- ReM: draw $\mathbf{Z}$ from CRE and accept it if and only if $M \leq a$ for some pre-specified a
 - $a = \infty$: CRE
 - $a = 0$: very few feasible treatment allocations
 - typical choices: $a = 0.001$ or some upper quantile of χ_K^2

Statistical inference

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- Neymanian inference: distribution of $\hat{\tau}$ given $M \leq a$

$$\frac{\hat{\tau} - \tau}{\sqrt{\text{var}(\hat{\tau})}} \stackrel{\cdot}{\sim} \sqrt{R^2} L_{K,a} + \sqrt{1 - R^2} \epsilon, \quad R^2 = \text{corr}^2(\hat{\tau}, \hat{\tau}_X)$$
$$L_{K,a} \sim D_1 \mid \mathbf{D}^\top \mathbf{D} \leq a, \quad \mathbf{D} = (D_1, \dots, D_K) \sim N_K(0, I_K)$$
$$\epsilon \sim N(0, 1), \quad L_{K,a} \perp\!\!\!\perp \epsilon$$

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Covariate adjustment

- How do we use covariates to improve the analysis
- Covariate-adjusted FRT
- Covariate-adjusted Neymanian inference

Covariate-adjusted FRT

- Test statistic with covariates: $T = T(\mathbf{Z}, \mathbf{Y}, \mathbf{X})$
 - pseudo-outcome strategy: use residuals from fitted models as outcome
 - model-output strategy: use model outputs as test statistic

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- We can use any model for the above two strategies

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 - coefficient of Z_i : $\hat{\tau}_F$
 - Fisher suggested using $\hat{\tau}_F$ as an estimate of τ to improve efficiency
- Freedman's critique
 - $\hat{\tau}_F$ is biased, while $\hat{\tau}$ is unbiased
 - $\hat{\tau}_F$ may not improve the efficiency over $\hat{\tau}$
 - standard error from the OLS is inconsistent for $\text{var}(\hat{\tau}_F)$ under CRE

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- Lin (2013)'s response as a Ph.D. thesis: you didn't do it right

Lin's estimator

- Consider a family of estimator

$$\begin{aligned}\hat{\tau}(\beta_1, \beta_0) &= n_1^{-1} \sum_{i=1}^n Z_i (Y_i - \beta_1^\top X_i) - n_0^{-1} \sum_{i=1}^n (1 - Z_i) (Y_i - \beta_0^\top X_i) \\ &= \left\{ \hat{Y}(1) - \beta_1^\top \hat{X}(1) \right\} - \left\{ \hat{Y}(0) - \beta_0^\top \hat{X}(0) \right\}\end{aligned}$$

- $\mathbb{E}\{\hat{\tau}(\beta_1, \beta_0)\} = \tau$ for any β_1, β_0
- $\hat{\tau}(\beta_1, \beta_0)$ is the difference in means of adjusted potential outcomes $\{Y_i(1) - \beta_1^\top X_i, Y_i(0) - \beta_0^\top X_i\}_{i=1}^n$

Lin's estimator

- Variance and variance estimator of $\hat{\tau}(\beta_1, \beta_0)$

$$\text{var}\{\hat{\tau}(\beta_1, \beta_0)\} = \frac{S^2(1; \beta_1)}{n_1} + \frac{S^2(0; \beta_0)}{n_0} - \frac{S^2(\tau; \beta_1, \beta_0)}{n}$$

$$\hat{\text{var}}\{\hat{\tau}(\beta_1, \beta_0)\} = \frac{\hat{S}^2(1; \beta_1)}{n_1} + \frac{\hat{S}^2(0; \beta_0)}{n_0}$$

$$\hat{S}^2(1; \beta_1) = (n_1 - 1)^{-1} \sum_{i=1}^n Z_i \left\{ (Y_i - \hat{Y}(1)) - \beta_1^\top (X_i - \hat{X}(1)) \right\}^2$$

$$\hat{S}^2(0; \beta_0) = (n_0 - 1)^{-1} \sum_{i=1}^n (1 - Z_i) \left\{ (Y_i - \hat{Y}(0)) - \beta_0^\top (X_i - \hat{X}(0)) \right\}^2$$

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- We can find the best (β_1, β_0) to
 - minimize $\hat{\text{var}}\{\hat{\tau}(\beta_1, \beta_0)\}$
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 - minimize $\hat{\text{var}}\{\hat{\tau}(\beta_1, \beta_0)\}$
 - minimize $\text{var}\{\hat{\tau}(\beta_1, \beta_0)\}$
 - the obtained estimator is the same

Lin's estimator

- Minimize $\text{var}\{\hat{\tau}(\beta_1, \beta_0)\}$

$$\hat{\beta}_1 = \underset{\beta_1}{\operatorname{argmin}} \sum_{i=1}^n Z_i \left\{ (Y_i - \hat{Y}(1)) - \beta_1^\top (X_i - \hat{X}(1)) \right\}^2$$

$$\hat{\beta}_0 = \underset{\beta_0}{\operatorname{argmin}} \sum_{i=1}^n (1 - Z_i) \left\{ (Y_i - \hat{Y}(0)) - \beta_0^\top (X_i - \hat{X}(0)) \right\}^2$$

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- $\hat{\beta}_z$ is the coefficient of X_i in the OLS fit of Y_i on $(1, X_i)$ for units with $Z_i = z$
- final estimator $\hat{\tau}(\hat{\beta}_1, \hat{\beta}_0)$

Lin's estimator

- Final estimator

$$\hat{\tau}(\hat{\beta}_1, \hat{\beta}_0) = \left\{ \hat{Y}(1) - \hat{\beta}_1^\top \hat{X}(1) \right\} - \left\{ \hat{Y}(0) - \hat{\beta}_0^\top \hat{X}(0) \right\}$$

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- define $\hat{\gamma}_z$ as the intercept in the OLS fit of Y_i on $(1, X_i)$ for units with $Z_i = z$
- $\hat{\gamma}_z = \hat{Y}(z) - \hat{\beta}_z^\top \hat{X}(z)$

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- $\hat{\tau}(\hat{\beta}_1, \hat{\beta}_0)$ is equal to the coefficient of Z_i in the OLS fit of Y_i on $(1, Z_i, X_i, Z_i X_i)$

Lin's estimator

- Variance estimator

$$\begin{aligned}\text{var}\{\hat{\tau}(\hat{\beta}_1, \hat{\beta}_0)\} &= \frac{1}{n_1(n_1 - 1)} \sum_{i=1}^n Z_i \left(Y_i - \hat{\gamma}_1 - \hat{\beta}_1^\top X_i \right)^2 \\ &\quad + \frac{1}{n_0(n_0 - 1)} \sum_{i=1}^n (1 - Z_i) \left(Y_i - \hat{\gamma}_0 - \hat{\beta}_0^\top X_i \right)^2\end{aligned}$$

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- Lin (2013) shows that the EHW standard error from the OLS of Y_i on $(1, Z_i, X_i, Z_i X_i)$ is almost identical to $\text{var}\{\hat{\tau}(\beta_1, \beta_0)\}$

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- Variance estimator

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- subtle issue of variance estimation

- Answer to Freedman's critique
 - include interaction terms in the regression
 - use EHW standard error
- Lin's estimator does not rely on correct linear model
 - OLS is an estimation strategy to obtain $\hat{\tau}(\beta_1, \beta_0)$
 - use OLS as long as τ is the target

Reducing Transphobia (Brookman and Kalla, 2016, *Science*)

- Can perspective-taking – “imagining the world from another’s perspective” – reduce transphobia?
- Treatment group ($n_1 = 912$): canvasser asking people to think about a time when they were judged unfairly and were guided to translate that experience to a transgender individual’s experience
- Control group ($n_0 = 913$): canvasser asking people to recycle
- Outcome variable: feeling thermometer towards transgender people (0 – 100)
 - ① baseline
 - ② 3 day
 - ③ 3 week
 - ④ 6 week
 - ⑤ 3 month

Durable Effects

- 6 week effect
- Dropping additional observations with missing covariates $n_1 = 278$ and $n_0 = 287$
- Difference in means: est. = 5.99, s.e. = 2.29, 95% CI = [1.50, 10.48]
- Simple regression: est. = 5.99
 - s.e. under homoscedasticity: 2.29
 - s.e. without homoscedasticity: 2.29
- Regression with covariates (age, gender, party ID, baseline thermometer rating)
 - no interaction: est. = 4.28, s.e. = 1.64 (HC)
 - with interaction: est. = 4.28, s.e. = 1.64 (HC)

Predictive and projective estimators

- Predictive estimator based on OLS fits
 - predict missing potential outcomes using linear models

$$\begin{aligned}\hat{\mu}_1(X_i) &= \hat{\gamma}_1 + \hat{\beta}_1^\top X_i, & \hat{\mu}_0(X_i) &= \hat{\gamma}_0 + \hat{\beta}_0^\top X_i \\ \hat{\tau}_{\text{pred}} &= n^{-1} \left\{ \sum_{Z_i=1} Y_i + \sum_{Z_i=0} \hat{\mu}_1(X_i) - \sum_{Z_i=1} \hat{\mu}_0(X_i) - \sum_{Z_i=1} Y_i \right\}\end{aligned}$$

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- $\hat{\tau}_{\text{pred}} = \hat{\tau}_{\text{proj}} = \hat{\tau}(\hat{\beta}_1, \hat{\beta}_0)$

Adjusting for covariate imbalance

- $\hat{\tau}(\beta_1, \beta_0) = \hat{\tau} - \beta^\top \hat{\tau}_X$, $\beta = \frac{n_0}{n} \beta_1 + \frac{n_1}{n} \beta_0$
 - Lin's estimator: $\hat{\tau}(\hat{\beta}_1, \hat{\beta}_0) = \hat{\tau} - \hat{\beta}^\top \hat{\tau}_X$
 - large covariate imbalance $\rightsquigarrow$ large difference between $\hat{\tau}$ and $\hat{\tau}(\hat{\beta}_1, \hat{\beta}_0)$

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- Intuition about efficiency gain: $\hat{\tau}$ and $\hat{\tau}_X$ are correlated

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 - large covariate imbalance $\rightsquigarrow$ large difference between $\hat{\tau}$ and $\hat{\tau}(\hat{\beta}_1, \hat{\beta}_0)$
- Intuition about efficiency gain: $\hat{\tau}$ and $\hat{\tau}_X$ are correlated
- Difference between $\hat{\tau}$ and $\hat{\tau}(\hat{\beta}_1, \hat{\beta}_0)$ depends on covariate balance
 - what if in practice we find $\hat{\tau}$ and $\hat{\tau}(\hat{\beta}_1, \hat{\beta}_0)$ very different?
 - different choices of $X_i \rightsquigarrow$ p-hacking
 - re-randomization $\rightsquigarrow$ reduce covariate imbalance

More remarks

- Covariate adjustment is not unique
 - other models can be used for obtaining $\hat{\mu}_1(X_i)$ and $\hat{\mu}_0(X_i)$
 - $\hat{\tau}(\beta_1, \beta_0)$ depends only on $\beta = \frac{n_0}{n}\beta_1 + \frac{n_1}{n}\beta_0$
- Extension: covariate adjustment in SRE
- Duality between ReM and covariate adjustment

Summary

- Simple linear regression $\rightsquigarrow$ Difference-in-means estimator
- Homoskedasticity implies the equal variance of potential outcomes
- Heteroskedasticity-robust variance estimator relaxes this assumption
- Covariate adjustment can be used to improve efficiency in randomized experiments
- Randomization of treatment assignment protects researchers from misspecification
 - linear regression estimators are valid without modeling assumption
- Blocking (Stratification) improves efficiency of inference with randomized experiments while preserving the advantages of randomization

For practitioners

- Regression method for randomized experiments:
 - without covariates: simple linear regression with robust variance
 - with covariates:
 - 1 calculate centered covariates
 - 2 run linear regression with interaction terms
 - 3 use robust variance
- Regression method for stratified randomized experiments:
 - without covariates:
 - 1 run linear regression with dummy variables for the stratification variable, include interaction terms
 - 2 calculate point estimator as the weighted average of the coefficients in the regression
 - 3 calculate variance estimator as the the weighted average of the robust variances
 - with covariates

Design versus statistical modeling

- From investigators' perspective
 - “if your experiment needs statistics, you ought to have done a better experiment” — Ernest Rutherford
 - “regression models often seem to be used to compensate for problems in measurement, data collection, and study design. By the time models are deployed, the scientific position is nearly hopeless” — David Freedman

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- From statistician's perspective
 - complications in handling humans afford a gold mine of interesting research questions
 - core knowledge and skills highly transferrable
 - statisticians get to play in everyone's backyard

Suggested readings

- Simple linear regression
 - ANGRIST AND PISCHKE, Chapter 2
 - IMBENS AND RUBIN, Chapter 7
- Covariate adjustment
 - IMBENS AND RUBIN, Chapters 7, 9
 - ANGRIST AND PISCHKE, Chapters 3.1
- Stratified experiments
 - IMBENS AND RUBIN, Chapter 9